

THATO BARUTI

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ML · Physics-Informed Learning · Surrogate Modelling · CFD Validation

Scientific

PROFILE

Scientific ML engineer completing an MSc in Computational Fluid Dynamics at Cranfield University, building graph-based, physics-informed, and hybrid surrogate models for aerodynamic design under real engineering constraints. My work spans dataset construction, graph learning, uncertainty calibration, multi-fidelity correction against higher-fidelity CFD, failure-regime auditing, optimisation, and deployment-facing reporting. I am strongest where machine learning, numerical methods, and engineering judgment meet: building models that are not only accurate, but also traceable, stress-tested, and usable in real simulation-driven surrogate and digital-twin-style workflows.

SELECTED IMPACT

- **Built a full aerodynamic surrogate stack in PyTorch and PyTorch Geometric:** developed DGCNN, MeshGraphNet-style, polar-structured, and target-wise hybrid models for lift, drag, and pitching-moment prediction across wing design sweeps.
- **Established evidence-grade evaluation instead of accuracy-only reporting:** implemented geometry-disjoint case-level splits, metadata baselines, strict compare gates, slice metrics, split-conformal uncertainty calibration, model-audit tools, and reliability ranking.
- **Delivered strong held-out performance with clear architectural complementarity:** MeshGraphNet-style model reached 3.18% for C_L , 2.60% for C_D , and 3.83% for C_{My} ; the polar-structured model improved C_D to 2.49%; the target-wise champion ensemble achieved 2.87% deployment-weighted held-out MAPE.

RESEARCH AND ENGINEERING EXPERIENCE

AeroGenic Scientific ML Platform

2025 – Ongoing

MSc Research, Cranfield University

- Built graph-based and physics-structured aerodynamic surrogate models in PyTorch and PyTorch Geometric, including DGCNN, MeshGraphNet-style message passing, FiLM conditioning, hybrid residual heads, and polar-law structure for interpretable coefficient prediction.
- Implemented the data and evaluation spine around the models: case-level geometry-disjoint train/validation/test splits, static k-NN graph building, metadata-only baselines, residual CSV generation, parity plots, regime-slice diagnostics, and all-model publication figures.
- Built uncertainty and audit tooling with split-conformal calibration, validation-case selection, and a lightweight reliability meta-model; locked-model audit proved 210/45/45 unique cases with zero overlap and achieved Spearman 0.613 between predicted and realised held-out risk.

Multi-Fidelity CFD Validation and Residual Correction

2025 – Ongoing

Cranfield University HPC Infrastructure

- Commissioned an automated OpenFOAM RANS validation pipeline and resolved structural failures across mesh generation, parallel execution, solver stability, function-object compatibility, and remote-rank environment setup by encoding the fixes directly into the case generator.
- Managed a 32-case validation campaign with 29 returned CFD cases and 26 strict-pass evidence cases, using the results to expose the viscous drag gap inherited from low-fidelity VLM-style labels and to prioritise the next validation regimes.
- Built grouped geometry-held-out residual correction layers against the higher-fidelity CFD evidence, reducing corrected MAE to 0.0677 for C_L , 0.0104 for C_D , and 0.401 for C_{My} .

Optimisation, Selection, and Deployment-Facing Tools

2025 – Ongoing

Cranfield University

- Implemented a requirement-conditioned Pareto optimisation gate over 63,393 candidate designs, retained 15,031 feasible lift-band candidates, and extracted 1,012 non-dominated solutions across drag, trim, lift overshoot, span, and training-manifold risk objectives.
- Built auditable candidate-selection tools that combine residual size, conformal interval width, aerodynamic stress variables, airfoil coverage, and representative anchor cases to decide which designs should receive higher-fidelity CFD validation next.
- Exposed the stack through deployment-facing FastAPI and Gradio interfaces that turn geometry and flight requirements into traceable wing-design outputs, confidence scores, exported geometry, and JSON reports.

Physics-Informed and Data-Driven ML Projects

2025

Cranfield University

- Built PINN surrogate models for 2D aerofoil flow in PyTorch with custom Navier–Stokes residual losses and bounded residual corrections, replacing expensive solver calls at inference time.
- Trained a FitNet model on high-dimensional DNS turbulence data, owning extraction, normalisation, train/validation logic, convergence monitoring, and error analysis against held-out ground truth; awarded 87% in Data Analysis & Post-Processing.
- Ran GPU-native Lattice Boltzmann simulations in FluidX3D and validated cylinder-flow and vortex-shedding behaviour against reference data, strengthening my understanding of GPU-bound simulation workloads and solver verification.

EDUCATION

MSc Computational Fluid Dynamics
Cranfield University, UK

Oct 2025 – Aug 2026 (Expected)

On track for distinction; selected modules include Data Analysis & Post-Processing (87%), Introduction to Fluid Mechanics & Heat Transfer (87%), Turbulence Modelling, Numerical Methods & HPC, Grid Generation, and Numerical Modelling.

BEng Aerospace Engineering
University of Sheffield, UK

Jun 2025

Awards: University of Sheffield International Undergraduate Scholarship; Botswana Government Top Achievers Scholarship.

ADDITIONAL ENGINEERING EXPERIENCE

AeroMAP UAS Build
Performance Lead, University of Sheffield

Oct 2023 – May 2024

Directed simulation-driven design optimisation across five components, validating against physical flight test and delivering +25% stability and +15% endurance.

UKSEDS National Rocketry Competition
Design Team Lead, University of Sheffield

Oct 2022 – May 2023

Led aerodynamic analysis and design iteration, reduced drag coefficient by 22%, and ranked Top 10 nationally among 30+ UK university teams.

TECHNICAL SKILLS

ML and Data	PyTorch; PyTorch Geometric; NumPy; scikit-learn; custom training loops; graph neural networks; surrogate modelling; uncertainty calibration; residual correction; Pareto optimisation
Architectures	DGCNN; MeshGraphNet-style message passing; PINNs; hybrid residual models; FiLM conditioning; polar-structured coefficient models
Engineering	Python; C++; Git; experiment evaluation; slice diagnostics; auditable reporting; FastAPI; Gradio
HPC and CFD	Linux HPC workflows; OpenFOAM; ANSYS Fluent; UCNS3D; PBS; SLURM; MPI-parallel simulation pipelines; ParaView; Tecplot; FluidX3D
Domain	Aerodynamics; CFD validation; wing design; simulation-driven optimisation; surrogate and digital-twin-style engineering AI workflows